

Description	Owners/Teams	Impact	Effort (FTE)	Milestones (M) and Decision Points (D) (year: description)	Risks and External Dependencies	Prob. of Applicability /Success/Adoption (Low, Medium, High)
Simulation						
2.10 Full G4-based application: Integration on CMSSW to run on GPUs and HPCs	Development team: community-wide including CERN, ANL/FNAL/Oakridge CMS ownership: Simulation (Vladimir Ivantchenko, Sam Bein + Kevin Pedro)	speedup G4 simulation applications per node by a large factor (2-10)	1.5 1.5	'22 D: community decision to move forward w/ GPU products '23-'25 M: Integration in CMSSW of first GPU prototypes '25-'26 M: workflows and tests for HPCs '26-'28 M: CMSSW integration, testing, validation of production-level version '28 D: decision on adoption for HL-LHC	Dependencies: community developed G4 version for GPUs in HPCs Risks: bottlenecks limiting overall speedup, lack of effort for <u>production-level</u> version, lack of effort for CMS integration	Adoption: Simu L2s (and K. Pedro) say Medium (L1s say Low)
Reconstruction						
2.22 Reconstruction: GPU-targeted pixel algorithm (Patatrack)	Development team, CMS ownership: Tracking POG, O&C reconstruction (J. Pata, C. Caputo), Marco Musich	5% of reco per-event time. (Target CPU-to-GPU ratio of ~2.)	1.0 1.0 1.0	'22 M: CUDA porting of pixel-only algorithm for use in HLT completed '22-'23 M: explore use of GPU version in offline reconstruction '23 D: Decision to port pixel offline algorithm to GPU '24-'25 M: Develop and produce first offline version (sing '26-'28 M: improvements, integration and validation of production-level version	Dependencies: cellular automation track-building Risks: SoA data structure sizes, CMS GPU/Portability framework.	Adoption: High
2.27 Reconstruction: GPU-targeted cellular automation track-building	Development team, CMS ownership: Tracking POG, reconstruction (J. Pata, C. Caputo), Marco Musich	10% of reco per-event time. (Reduce by offloading to GPU.)	1.0 1.0 1.0	'23 M: algorithm implemented for high-pT prompt tracks '24 D: verify, improve physics performance (acceptable @140/200 PU)	Dependencies: portability tool, track seeding (downstream)	Success: Medium
[No updates from experts since LHCC Review]			1.0	'25-'26 M: optimization, expansion to use outer tracker hits '26 M: validation of production-level version		
2.14 Reconstruction: GPU-targeted electron seeding algorithms	Development team, CMS ownership: Egamma POG, reconstruction (J. Pata, C. Caputo)	9% of reco per-event time. (Reduce by offloading to GPU.)	0.5	'23 M: Complete prototype. '24 D: Decision to move forward for development	Dependencies: portability tool Risks: Patatrack doublets are not saved	Success: Medium
2.19 Reconstruction: GPU-targeted HGCal reco with TICL	Development team, CMS ownership: HGCal DPG, reconstruction (J. Pata, C. Caputo)	5% of reco per-event time. (Reduce orders of magnitude by offloading to GPU.)	2.0	'26 M: full reconstruction, including high-level showers implemented to run on GPU	Dependencies: portability tool, TensorFlow/Pythorch/ onnx Risks: current TICL algos and data structures not optimal for parallel computing, insufficient effort	Success: High
2.21 Reconstruction: HGCal ML reconstruction	Development team, CMS ownership: HGCal DPG, reconstruction (J. Pata, C. Caputo), ML (J.R. Vlimant, G. Kasieczka, J. Ngadiuba), L. Gray	5% of reco per-event time. (Improve physics and reduce orders of magnitude by offloading to GPU.)	2.0	'22 M: train ML algorithm for multi-particles in low pileup '23-'25 M: multi-particles algorithm running on high pileup environment, HGCal workflow integrated in CMSSW and inference as a service infrastructure. '25 D: decide on viability of approach for HL-LHC '26-'28 M: improve algorithm, integration, validation, deliver production-level version	Dependencies: portability tool, TensorFlow/Pythorch/ onnx Risks: viability of large-scale GNN in CMSSW, achieve competitive physics performance	Success: Medium
2.20 Reconstruction: GPU-targeted	Development team, CMS ownership: ML, ECAL, HCAL,	7% of reco per-event time.	1.5	'22 (M): Simulation-based training '23 (M): Test improvements over	Dependencies: portability tools, TensorFlow/Pythorch/	Success: Medium
Particle Flow reconstruction with ML	PF, JME, reconstruction (J. Pata, C. Caputo), ML (J.R. Vlimant, G. Kasieczka, J. Ngadiuba)	(Improve physics and reduce orders of magnitude by offloading to GPU.)		standard PF '24 (M): Standalone tests using Run3 data subsets. '25 (D): Decision to develop a production level version for HL-LHC	onnx Risks: viability of large-scale GNN in CMSSW, achieve competitive physics performance	